

FROM TRACES TO GUARDED PROGRAMS: EVIDENCE-GATED COMPILATION OF RECURRENT AGENT WORKFLOWS

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ABSTRACT

Tool-using agents often revisit the same read-only evidence path while paying for a model decision at every tool boundary. Replacing a recurring model-mediated prefix is tempting, but not ordinary caching. A later tool argument may encode a model decision that cannot be recovered from observed state; an apparently read-only call may cross an effect boundary; and a program that replays tool calls correctly may still change the answer produced by the remaining agent. The central question is therefore not whether a trace repeats, but whether the available evidence justifies substituting the model.

We introduce guarded agentic compaction (GAC), a trace-to-program compiler for this decision. GAC normalizes recorded episodes, mines recurrent read-only prefixes, and admits a region only if *every* tool argument in it can be reconstructed from entry state or a prior result; one unwitnessed argument retires the region, leaving the maximal justified prefix. Surviving regions become bounded deterministic programs with manifest, input, and output guards, and deploy only when a finite-sample selective-risk test certifies their error bound; otherwise GAC retires the candidate and runs the unchanged agent. On three live GitHub workflow families, GAC reaches 90/90 exact held-out contracts against 89/90 for the unchanged agent while reducing provider requests by 66.6% and tokens by 63.1% in aggregate. On a time-forward five-repository extension it completes four repositories and retires one; on NESTFUL and API-Bank it retires every recurrent family despite successful provenance, synthesis, and replay. Thus recurrence identifies a candidate, whereas evidence determines whether replacing the model is justified.

1 INTRODUCTION

The standard agent loop pays for a model decision at every tool boundary: model, tool, model, tool, and so on (Yao et al., 2023). That flexibility is valuable while a workflow is novel. At scale, however, mature agents often revisit the same read-only evidence path with the same data dependencies. In those settings, repeating the model decision adds latency, tokens, and cost without necessarily adding useful adaptation.

Removing that decision is not ordinary caching. A tool argument may depend on a specific earlier observation, a repeated sequence may cross an approval or write, an apparently read-only tool may hide an undeclared effect, and a program that replays the same tool outputs may still change the downstream answer. The problem is therefore not just “find recurring traces.” The problem is to decide when a recurrent model-mediated region can be replaced by a deterministic program and when it should be refused.

Consider held-out issue #4420 in the expanded 30-record study. Given only `issue_number=4420`, the observed trace calls `record(4420)` and then passes the *returned* `record.issue_number` to `labels` and to `comments`, which it invokes with `limit=100` before rendering an auditable triage answer. All 132 discovery executions take that route, so a recurrence-only optimizer would emit all three reads. GAC instead retires the *region*: `comments.limit` varies across otherwise similar entries and is reachable neither from the entry state nor from either earlier result, so one unwitnessed slot blocks the whole three-read candidate as `ungroundable_slot`. What survives is the maximal justified prefix `record → labels`; the agent still chooses the comment limit and writes the answer. That two-read artifact passes 92/92 calibration groups at $\alpha = .05$ (upper bound 0.0498).

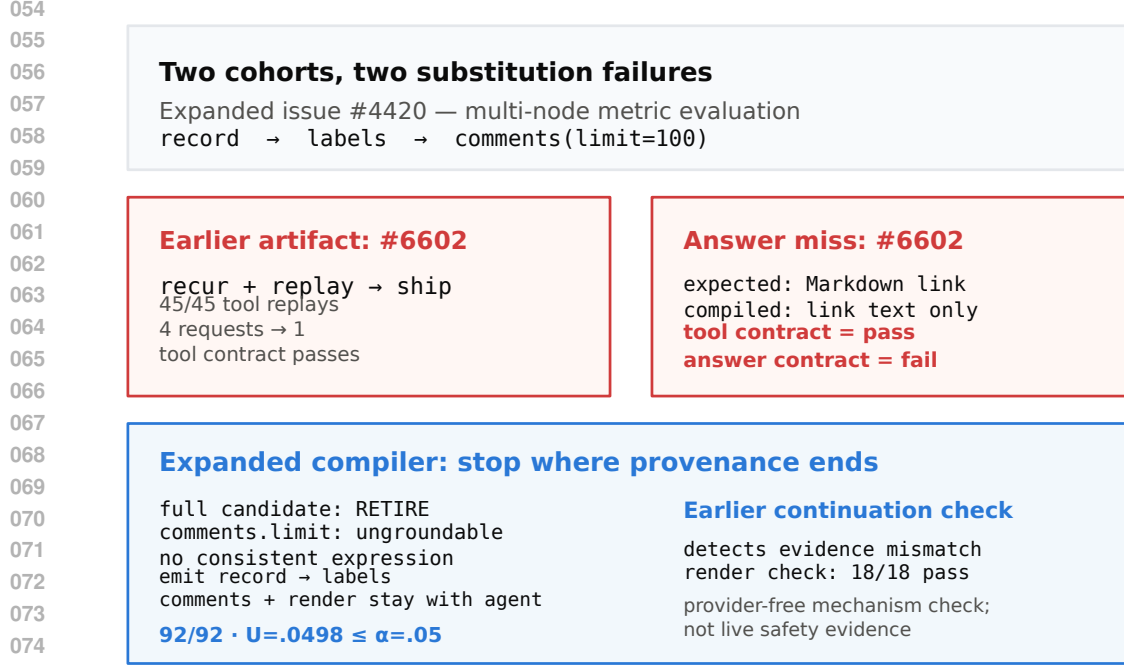


Figure 1: Two retained cohorts expose different failure modes. In the expanded study, issue #4420 forces a shorter grounded prefix; in the earlier study, issue #6602 shows that clean tool replay need not preserve the downstream answer.

Why stop there? A separate, earlier 18-record study supplies the counterexample: its aggressive three-read artifact replayed all 45 tool calls, yet issue #6602 lost the URL from a Markdown link and only 17/18 downstream answers remained exact. A provider-free continuation guard detects and checked-renders that miss; it is a mechanism check, not live safety evidence. Together the two cohorts (fig. 1) show why recurrence proposes a candidate, while provenance, continuation scope, and statistical evidence determine what may replace the model.

Framed this way, the closest precedent is not prompt optimization but the tracing just-in-time compiler. Dynamo (Bala et al., 2000) detects a hot path, compiles it behind a *guard*, and deoptimizes to the interpreter when the guard fails (Hölzle et al., 1992). Agent traces admit the same structure — hot-trace detection becomes family mining, guard synthesis becomes an entry contract, and deoptimization becomes fallback to the unchanged agent — but they add three obligations a JIT does not face. Tool arguments must be shown to be *grounded* in observable state rather than invented by the model; declared *effects* must permit speculation, because an agent’s “interpreter” can call the outside world; and the residual error rate of a compiled region is statistical, so it needs a finite-sample bound rather than a soundness proof.

We study these obligations through *guarded agentic compaction* (GAC), a compile-or-retire trace-to-program system for recurrent read-only prefixes (fig. 2). The core contribution is not “agent compilation” in general, but an admissibility argument for trace-derived specialization:

1. A guarded specialization pipeline (Alg. 1) that combines typed provenance, effect and position barriers, bounded synthesis, runtime verification, and finite-sample compile-or-retire admission (Alg. 2, proposition 1); every stage can refuse, and the default output is the unchanged agent.
2. A real-provider evaluation across three public-record workflow families, plus a narrower cross-repository, time-forward extension that keeps both successful artifacts and principled retirements.
3. A negative-result boundary showing that recurrence and replayability do not themselves license deployment: NESTFUL and API-Bank retire under the same admission logic, and hand-written

programs are competitive or tied wherever the workflow is already known, so no claim of automatic superiority is warranted.

2 PROBLEM FORMULATION

Episodes and candidate regions. An episode $E = (z, M, e_{1:T}, y)$ contains an entry-state snapshot z , a content-addressed execution manifest M , ordered events e_t (model requests and responses, tool calls and results, handoffs, guardrails, approvals, errors, commit boundaries), and an observable outcome y . Each tool f carries an application-owned effect declaration $\epsilon(f)$ and capability flags such as *speculatable*; an unknown effect is not a read.

A candidate region $R = [a, b]$ begins at a model-request boundary and ends after a tool result. It is *groundable* when every call argument is an expression over entry state or prior results inside R ,

$$\forall c_j \in R, \forall u \in \text{args}(c_j) : u = g_{j,u}(z, o_{<j}), \quad g_{j,u} \in \mathcal{L}, \quad (1)$$

for a closed, bounded transform library $\mathcal{L}$; and *effect-admissible* when every tool in R is declared read-only, speculatable, replayable, and inside one principal and isolation partition. Handoffs, approvals, writes, errors, and unknown effects are hard barriers, not costs to be traded away. Because the deployed runtime resolves artifacts at the initial model boundary, the compiler also imposes a *position invariant*:

$$a = 0 \quad \text{in the normalized model-boundary index.} \quad (2)$$

This is not aesthetic. A suffix program dispatched at entry can reorder or duplicate calls the agent has already made — a semantic mismatch between compilation-time and resolution-time position rather than a tuning failure, and a pilot that violated eq. (2) produced exactly that fault (appendix E).

Selective optimization objective. The compiler emits an artifact $A = (P, H, V, q, \eta, M)$: a deterministic program P , hard guard H , output verifier V , nonconformity score q , threshold η , and manifest pins M . On a future episode with entry state z , dispatch is *selective*,

$$d_A(z) = \mathbf{1}\{H(z) = 1 \wedge q(z) \leq \eta \wedge M \text{ matches}\}, \quad (3)$$

and any failed precondition runs the unchanged baseline agent B . If execution fails before a commit and staging proves the attempt clean, B runs; a dirty failure after a commit is an incident, not a fallback. Writing $L(A, z) \in \{0, 1\}$ for a wrong compiled execution *after* dispatch and $C(\cdot, z)$ for execution cost, we seek

$$\max_A \mathbb{E}[d_A(z) \{C(B, z) - C(A, z)\}] \quad \text{s.t.} \quad \Pr[L(A, z) = 1 \mid d_A(z) = 1] \leq \alpha. \quad (4)$$

Two properties shape everything that follows. The constraint conditions on dispatch, so abstention is free and only post-dispatch errors are charged — the reject-option formulation applied to a compiler rather than a classifier — and returning *no* artifact is feasible, so `RETIRE` is a legitimate output rather than a failure mode. Equation (4) is a design target: the implementation returns the first family that survives every stage of Alg. 1, and we prove no bound on the gap to the best one. Calibration uses independent scenario groups as its statistical units: a group is admitted if any member dispatches and violates if any dispatched member is wrong. This conservative group risk equals episode risk in the reported live studies, where each group is one public record.

3 GUARDED AGENTIC COMPACTION

Figure 2 and Alg. 1 give the same six stages at two levels of detail. The order is itself part of the argument: each stage can only weaken a candidate, so a later stage may reject one an earlier stage passed, but no later stage can authorize one an earlier stage blocked. We describe each and state what it can refuse.

1. Typed provenance. A capture adapter normalizes framework traces into the Episode IR; manifests pin prompt, policy, guardrail, tool, model, SDK, tracer, entry-contract, and effect-catalog identities, and episodes with different compatibility keys are partitioned before any learning. For each call-argument slot, the program-argument trace graph (PATG, Alg. 3) searches entry-state and

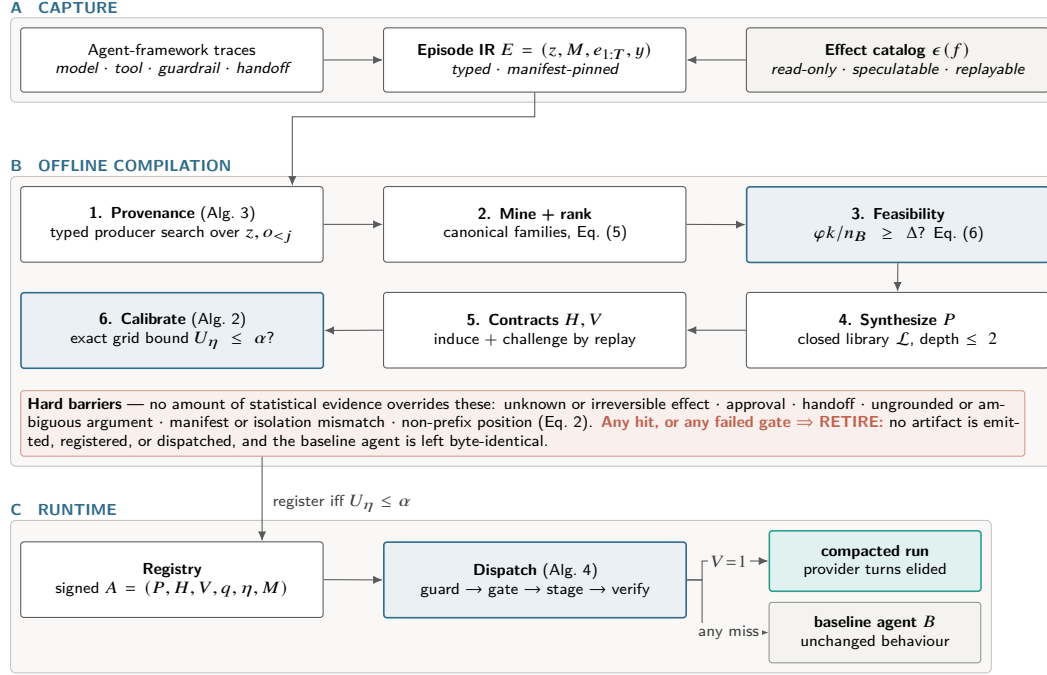


Figure 2: Architecture of GAC. (A) Capture normalizes traces into one typed Episode IR against a declared effect catalog. (B) Offline compilation is a cascade of independent rejection points; the shaded band lists barriers no statistical evidence can override. (C) At runtime exactly one terminal edge compacts; the rest return the unchanged agent.

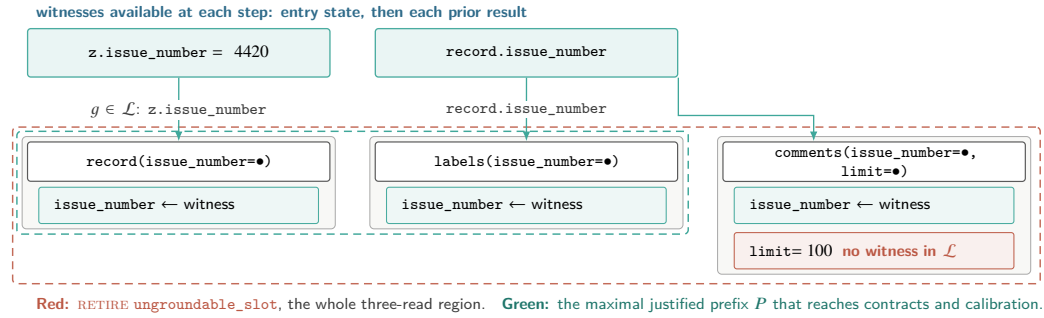


Figure 3: Equation (1) on the held-out issue-4420 record. Every argument slot needs an expression in $\mathcal{L}$ over the entry state or a prior result; the two issue_number slots have one and the limit slot has none, so it is drawn as a socket with no incoming edge. The question is strictly harder than recurrence: not “did this sequence repeat?” — all 132 discovery executions did — but “can every socket be wired from an allowed prior value?” One unwitnessed slot retires the whole region, and a slot with more than κ witnesses does the same.

prior-result paths for typed values reachable by a bounded transform — execution provenance in the database sense (Cheney et al., 2009), specialized to tool arguments. Two outcomes block a region rather than guessing: a slot with no witness is a genuine model decision, and a slot with more than $\kappa = 3$ witnesses is ambiguous. Figure 3 shows both branches on the motivating record. A repeated tool name, a matching literal, and a successful replay are not witnesses; only an expression in $\mathcal{L}$ is.

2. Region mining and ranking. The miner enumerates model-boundary-to-result windows of at most $w_{\max}$ tool calls, qualifies effects and live-ins, and hashes tool signatures, dependency topology,

Algorithm 1 COMPILER — the compile-or-retire cascade. Every exit is either a registered artifact or an attributed RETIRE, so rejections stay in the denominator. Stages marked (CALLER) are separate library entry points run in this order rather than steps inside one function.

Input: episodes $\mathcal{E}$; effect catalog C ; entry schema Θ ; budgets α, δ ; grid Λ ; feasibility target Δ

Output: artifact $A = (P, H, V, q, \eta, M)$, or RETIRE with an attributed reason

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1:  $\mathcal{E} \leftarrow \text{QUALIFY}(\mathcal{E}, C)$ ;  $\{\mathcal{E}_M\} \leftarrow \text{PARTITIONBY}(\mathcal{E}, \text{manifest}, \text{principal}, \text{isolation})$   $\triangleright$  drop partial runs,
   unpinned manifests
2: for all partitions  $\mathcal{E}_M$  do
3:    $(\mathcal{T}, \mathcal{D}, \mathcal{K}, S) \leftarrow \text{GROUPEDSPLIT}(\mathcal{E}_M)$  (CALLER)  $\triangleright$  train / dev / calibration / sealed test, split by group
4:    $G \leftarrow \text{BUILDPATG}(\mathcal{T}, \Theta)$   $\triangleright$  Alg. 3: block ungrounded or ambiguous slots
5:    $\mathcal{F} \leftarrow \text{CANONICALIZE}(\text{MINEREGIONS}(G, C, w_{\max}))$ , keeping only cross-group support
6:   if  $\varphi k / n_B < \Delta$  then (CALLER)
7:     record RETIRE (infeasible ceiling, Eq. 6); continue  $\triangleright$  try the next compatible partition
8:   end if
9:   for all families  $F \in \mathcal{F}$  ranked by Eq. (5) do
10:     $P \leftarrow \text{SYNTHESIZE}(F, \mathcal{L})$  at depth  $\leq 2$ ; if  $P = \perp$  then continue  $\triangleright$  ungroundable slot
11:     $(H, V) \leftarrow \text{INDUCECONTRACT}(F)$ ; if  $\neg \text{REPLAYANDCHALLENGE}(P, V, \mathcal{D})$  then continue  $\triangleright$ 
      counterexample
12:     $q \leftarrow \text{FITSCORE}(\mathcal{D})$ ; freeze  $q$   $\triangleright$  dev groups only; never calibration
13:     $\eta \leftarrow \text{CALIBRATE}(q, H, \mathcal{K}, \Lambda, \alpha, \delta)$ ; if  $\eta = \perp$  then continue  $\triangleright$  Alg. 2
14:    return  $\text{PACKAGE}(P, H, V, q, \eta, M)$  with evidence and lifecycle
15:  end for
16: end for
17: return RETIRE

```

run shape, and live-in/live-out shape while abstracting literal values. Families require support across independent scenario *groups* rather than raw event frequency, which prevents a single chatty scenario from manufacturing recurrence. Survivors are ranked by a minimum-description-length trade-off that pays for expected savings and charges for heterogeneity, effect exposure, and program size:

$$\text{score}(F) = \underbrace{s_F \bar{k}_F c_m}_{\text{expected saving}} - \lambda_1 H(F) - \lambda_2 \rho_{\text{eff}}(F) - \lambda_3 |F|, \quad (5)$$

with s_F the group support, $\bar{k}_F$ the mean removable model requests, c_m their unit cost, and $H(F)$ the Shannon entropy over canonical variants inside the family; without λ_1 the ranking prefers a heterogeneous family whose single canonical form fits none of its members. Equation (5) affects only *which* family is tried first — no admission decision depends on it.

3. Feasibility ceiling. Before any synthesis runs, an estimator bounds what the compiler could achieve with a perfect verifier and a gate that never abstains:

$$\Delta_{\max} = \frac{\varphi k}{n_B}, \quad \text{necessary: } \varphi \rho k \geq \Delta n_B, \quad (6)$$

with n_B baseline model requests per episode, φ the fraction of episodes holding an eligible region, k the requests removed per successful dispatch, and ρ the verifier pass rate. Since eq. (6) assumes $\rho = 1$, no measured reduction can exceed it, and it is met with equality exactly when nothing abstains and nothing fails. Reporting it first makes a decisive negative answer cheap: a workload whose ceiling sits under the target can be declined without building a compiler for it.

4–5. Bounded synthesis and contracts. Synthesis searches a 23-operator library at depth at most two, covering path selection, indexing, string normalization, arithmetic, comparisons, bounded collection operations, and narrow loops; bindings are ranked by stability across groups, and a group-wise refit rejects bindings that exploit row-level coincidence. The hypothesis space is deliberately small enough to read: the approach resembles programming-by-example (Gulwani, 2011) and bounded sketching (Solar-Lezama et al., 2006), and it does *not* generate arbitrary Python. Hard guards then constrain entry types, required fields, categorical sets, numeric intervals, patterns, isolation keys, allowed effects, and manifest pins; verifiers constrain live-out presence, type, cardinality, empirical hulls, provenance, effect multisets, and call counts. These are candidate invariants, not proofs for all future inputs — which is why a statistical gate follows them rather than replacing them.

Algorithm 2 CALIBRATE — fixed-grid exact selective admission. The score q and the grid Λ are frozen *before* this procedure sees a calibration group, which is what makes the union bound legitimate.

Input: frozen score q ; hard guard H ; calibration groups $\mathcal{K}$; grid Λ ; budgets α, δ

Output: threshold η with a simultaneous guarantee, or $\perp$ (RETIRE)

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1:  $\mathcal{A} \leftarrow \emptyset$  ▷ admissible (coverage, threshold) pairs
2: for all  $\eta \in \Lambda$  do
3:    $K_\eta \leftarrow \{g \in \mathcal{K} : \exists z \in g, q(z) \leq \eta \wedge H(z) = 1\}$ ;  $n_\eta \leftarrow |K_\eta|$ ; if  $n_\eta = 0$  then continue ▷ vacuous
4:   bound  $U_\eta = 1$ 
5:    $k_\eta \leftarrow |\{g \in K_\eta : g \text{ contains a violation}\}|$  ▷ wrong after dispatch only, never abstention
6:    $U_\eta \leftarrow 1$  if  $k_\eta = n_\eta$ , else  $\text{Beta}^{-1}(1 - \frac{\delta}{|\Lambda|}; k_\eta + 1, n_\eta - k_\eta)$ ; if  $U_\eta \leq \alpha$  then  $\mathcal{A} \leftarrow \mathcal{A} \cup \{(n_\eta/|\mathcal{K}|, \eta)\}$ 
7:   ▷ Eq. (7)
8: end for
9: return  $\perp$  if  $\mathcal{A} = \emptyset$ , else  $\arg \max_{(\text{cov}, \eta) \in \mathcal{A}} \text{cov}$  ▷ largest coverage among certified thresholds

```

6. Exact risk-gated admission. Algorithm 2 decides deployment. The score q is fitted on *dev* groups only — never on calibration groups — from entry-observable features such as missing fields, hull margin, temporal drift, and provenance ambiguity, then frozen together with the finite threshold grid $\Lambda = \{.02, .05, .08, .11, .14, .17, .20, .25, .30, .40, .50\}$. Two data signals do two different jobs here, and conflating them is the easiest way to misread the guarantee: q is *fitted* on development groups that turned out unproductive in any way — wrong or abstained — while the bound below is *counted* only from groups that were dispatched and turned out wrong. For each $\eta \in \Lambda$ the compiler counts admitted calibration groups (n_η) and those containing at least one post-dispatch violation (k_η), and inverts the one-sided exact binomial test (Clopper & Pearson, 1934) at a Bonferroni-corrected level, with explicit empty and all-violation cases:

$$U_\eta = \begin{cases} 1, & n_\eta = 0 \text{ or } k_\eta = n_\eta, \\ \text{Beta}^{-1}\left(1 - \frac{\delta}{|\Lambda|}; k_\eta + 1, n_\eta - k_\eta\right), & 0 \leq k_\eta < n_\eta, \end{cases} \quad U_\eta|_{k_\eta=0} = 1 - \left(\frac{\delta}{|\Lambda|}\right)^{1/n_\eta}. \quad (7)$$

It admits the largest-coverage threshold with $U_\eta \leq \alpha$ and retires the family otherwise — a fixed-grid instance of the learn-then-test discipline (Angelopoulos et al., 2022). The closed form on the right is the operative constraint in our experiments: at $\alpha = 0.05$, $\delta = 0.1$, and $|\Lambda| = 11$, a zero-violation family still needs $n_\eta \geq 92$ admitted groups. The default output is refusal, not emission.

Proposition 1 (Per-candidate selective-risk admission). *Fix one candidate program, hard guard, verifier, score, group-level violation rule, and finite grid Λ before calibration. If calibration groups are i.i.d. from the future-group distribution, then with probability at least $1 - \delta$, every $\eta \in \Lambda$ satisfies $r_\eta \leq U_\eta$, where r_η is post-dispatch group risk. Consequently the data-selected threshold $\hat{\eta}$ satisfies $r_{\hat{\eta}} \leq \alpha$ whenever $U_{\hat{\eta}} \leq \alpha$.*

Appendix A gives the proof. The guarantee is deliberately narrow: it is *per fixed candidate*, not compiler-wide over the adaptive candidate search Alg. 1 actually performs, and it conditions on group exchangeability rather than establishing it (section 7). The scientific point is that guarded deployment requires an explicit evidence boundary, not just a recurrent trace and a successful replay.

4 EXPERIMENTAL SETUP

Because the claim is an *admissibility* claim, the evaluation asks three questions: does specialization preserve outcomes while saving work on real records (section 5.1)? Does it transfer under a time-forward split (section 5.2)? And does it refuse when evidence is thin (section 5.3)?

4.1 BENCHMARK AT A GLANCE

Every GitHub sample begins with one issue or pull-request number from a pinned public snapshot. The agent must make read-only calls and return exact, source-grounded fields, not a subjective preference. The three primary families use 132 discovery and 30 balanced held-out records each, with live provider calls. Table 1 shows what one sample asks an agent to read and return; it tests both the structured answer and whether its evidence remains available after compaction.

Table 1: Primary GitHub benchmark at a glance. Each sample supplies one record number from a pinned public snapshot; every listed answer field is checked against the source record.

Task	Read-only evidence path	Exact answer contract
Issue type	<code>issue_number</code> → record, official labels, comments	Title, state, label-derived category, verbatim excerpt
PR outcome	<code>pr_number</code> → PR record, merge status, discussion	<code>open</code> , <code>merged</code> , or <code>closed_unmerged</code> , plus title and comment excerpt
Backlog attention	<code>issue_number</code> → open-issue record, assignees, discussion	<code>owned</code> , <code>discussed_unowned</code> , or <code>awaiting_first_response</code> , plus title, owner, and excerpt

Table 2: Primary live-provider results ($n = 30$ held-out records per family). “Interfaces” counts provider-visible tool interfaces. All conditions see identical records, model, and grader; *Manual* is hand-written with full knowledge of the workflow.

Family	Exact held-out contract			Reduction of compiled vs. baseline (%)				
	Base	Compiled	Manual	Requests	Interfaces	Tokens	Latency	Cost
Issue-type routing	30/30	30/30	30/30	50.0	0.0	39.5	51.7	32.0
PR-outcome audit	30/30	30/30	30/30	75.0	66.7	80.8	73.0	75.3
Backlog-attention routing	29/30	30/30	30/30	74.8	66.3	81.4	68.9	75.1
Weighted total ($n = 90$)	89/90	90/90	90/90	66.6	44.2	63.1	64.2	58.7

Cross-repository time-forward extension. A narrower two-read PR-outcome task spans five frozen public repositories. Held-out records are strictly newer than discovery records, and selection is sealed before provider calls; repositories without an admitted artifact remain in the result. A balanced rerun on three repositories controls the `open/merged/closed_unmerged` mix.

External refusal benchmarks. NESTFUL and API-Bank supply valid executable traces whose evidence is still insufficient; they retain the argument-level producer structure that provenance, synthesis, replay, and admission need, and they measure none of them as live planning quality (Basu et al., 2025; Li et al., 2023).

Conditions, metrics, and statistics. Each GitHub family compares the unchanged baseline B , the compiled artifact, and a hand-written pre-model program on the same records, so the comparison is paired rather than across cohorts. Quality is exact task-contract satisfaction; resources are requests, interfaces, tokens, latency, and estimated cost. We use paired exact McNemar tests, bootstrap intervals, and Wilcoxon signed-rank tests; the only guaranteed multiplicity control is the Bonferroni term in eq. (7). All admission runs use $\alpha = 0.05$, $\delta = 0.1$, and $|\Lambda| = 11$ (appendix C).

5 RESULTS

5.1 PRIMARY WORKFLOWS: PRESERVED CONTRACTS, LESS WORK

Across the 90 held-out records, GAC satisfies 90/90 exact contracts versus 89/90 for the unchanged agent (table 2). It reduces provider requests by 66.6%, tokens by 63.1%, latency by 64.2%, and estimated cost by 58.7% in aggregate. The per-family pattern is intuitive: issue-type routing admits only a two-read prefix, whereas the two deeper workflows approach 75% fewer requests.

No held-out record is a compiled-only failure; the one-sided 95% upper bound on that discordance is 3.3%. This supports preservation on the observed cohort, not equivalence in general. Each artifact was admitted with 92 zero-violation calibration groups ($U_\eta = 0.0498 \leq .05$). Hand-written programs also achieve 90/90 and are cheaper for issue-type routing, so the evidence supports automatic discovery and guarded lifecycle management—not runtime dominance over correct manual code.

Table 3: Cross-repository time-forward results on the narrower two-read pull-request outcome task. Held-out records are strictly newer than discovery records and selection is sealed before any provider call. The 5-repo cohort retires `pytorch/pytorch`. A fixed two-read template is more efficient in the 5-repo core (66.7% fewer requests against 44.4%) and essentially tied in the balanced rerun, which is why this result is evidence of automatic discovery and principled retirement rather than of runtime dominance.

Setting	Repos	Exact task contract			Reduction vs. baseline (%)			
		Discovery	Held-out	Classes	Req.	Tokens	Latency	Cost
5-repo core	4/5	580/580	120/120	skewed	44.4	52.4	49.4	48.6
3-repo rerun (balanced)	3/3	360/360	180/180	balanced	66.7	78.4	60.7	72.7

Table 4: Where refusal happens. Both benchmarks clear provenance, synthesis, and replay, then retire at calibration with *zero* wrong held-out executions: the binding constraint is sample size, not correctness. Stages refer to Alg. 1.

Stage of Alg. 1	NESTFUL	API-Bank
Episodes / recorded traces	1,415	212
Tools compilable / effect-blocked	—	21 / 28
Dependency slots with producer recovered (1)	5,531 / 5,746 (96.3%)	559 / 881 (63.5%)
Episodes yielding an admissible window (2)	1,207 (85.3%)	37 (17.5%)
Families reaching the support floor (2)	32	19
Families that synthesize a program (4)	12	2
Held-out replay: pass / abstain / wrong (5)	24 / 12 / 0	0 / 2 / 0
Largest family support $n_{\max}$ (6)	26	8
Groups required at (6), Eq. (7)	92	92
Artifacts admitted	0	0

5.2 TIME-FORWARD TRANSFER: COMPILE OR RETIRE

In the five-repository cohort, GAC completes 120/120 exact held-out pairs on four repositories and retires `pytorch/pytorch` at compile time rather than loosening its gate (table 3); the balanced three-repository rerun reaches 180/180 exact pairs with 66.7% fewer requests.

5.3 REFUSAL WHEN EVIDENCE IS INSUFFICIENT

The external benchmarks expose the opposite outcome. NESTFUL clears provenance, synthesis, and held-out replay: 12 families synthesize, with 24 replays passing and none wrong. Yet its largest family has only 26 groups, below the 92 zero-violation groups required by eq. (7), so every candidate retires (table 4). API-Bank fails earlier: its effect catalog blocks 28 of 49 tools before mining, leaving maximum support of eight; its two synthesized candidates also retire.

Both benchmarks therefore separate trace-level properties from deployment evidence: provenance reachability and replay correctness identify a plausible program without certifying that it should replace a model.

6 RELATED WORK

Guarded trace and agent workflow compilation. GAC adapts the tracing-JIT pattern—compile a hot path behind a guard and return to the interpreter on a miss (Bala et al., 2000; Hölzle et al., 1992)—to agents, whose arguments must remain grounded in observed state and whose speculation must be licensed by declared effects (Lucassen & Gifford, 1988). Recent systems mine meta-tools, induce workflows, cache plans, compile typed orchestration, validate tool-state plans, search workflow graphs, or prune multi-agent topology (Abuzakuk et al., 2026; Wang et al., 2025; Zhang et al., 2025; Wei et al., 2026; Winston et al., 2026; Li et al., 2026; Chen et al., 2026). They share our premise that recurrence is exploitable and differ in what they demand before exploiting it. The delta is three-fold. *First*, they take a recurring subsequence as the unit; we take the subsequence as a proposal and the

argument-level witness as the unit, which is why the motivating region shrinks from three reads to two instead of being accepted or dropped whole. *Second*, their guards test the input, whereas ours are a cascade in which provenance, effects, position, contracts, and a finite-sample bound each hold a veto — so a candidate can fail for reasons no additional trace evidence can repair. *Third*, they report what compilation achieved, whereas a compile-or-retire system must also report what it refused.

Other optimization targets. DSPy optimizes prompts, GEPA evolves them, RouteLLM changes models, and LLMCompiler parallelizes current calls (Khattab et al., 2024; Agrawal et al., 2026; Ong et al., 2025; Kim et al., 2024). GAC instead replaces a historical boundary only after guarded evidence.

Synthesis, risk control, and evaluation. The bounded search draws on programming by example, sketching, and execution provenance (Gulwani, 2011; Solar-Lezama et al., 2006; Cheney et al., 2009); admission uses a Clopper–Pearson bound with fixed-grid learn-then-test, not a new bound (Clopper & Pearson, 1934; Angelopoulos et al., 2022). BFCL and ToolLLM score tool use broadly (Patil et al., 2025; Qin et al., 2023); NESTFUL and API-Bank retain the producer references and call results needed for post-trace provenance, replay, and refusal (Basu et al., 2025; Li et al., 2023).

7 DISCUSSION AND LIMITATIONS

The strongest supported claim is narrow: repeated read-only workflow prefixes can become guarded deterministic programs that preserve exact held-out contracts on real GitHub families while failing closed when evidence is inadequate, including on NESTFUL, API-Bank, and one cross-repository cohort.

Gate maturity is the main scientific risk. Proposition 1 is per-candidate, not compiler-wide over the adaptive search of Alg. 1; certification requires freezing the candidate relative to calibration, and a Bonferroni repair over m candidates would use $\gamma = \delta/(m|\Lambda|)$, raising the zero-violation requirement from 92 admitted groups to 106 at $m = 2$. The gate is also empirically step-like at $n_\eta \geq 92$: we show that it *refuses* correctly far better than that q *discriminates*. Refitting q for a genuine frontier is the key follow-up.

Group independence is assumed, not established. Proposition 1 needs i.i.d. groups. The live studies set `min_days=1` and `min_principals=1` over one repository snapshot, so a “group” is one record from one project at one instant, sharing a schema, an author community, and a period with its neighbours. The grouped split makes exchangeability plausible; nothing in the calibration verifies it.

External validity and practical competitiveness. The richer evidence comes from one revision-pinned repository and one provider/model family, and the cross-repository extension is a narrower, template-tied task. We establish no superiority under drift or across providers, and do not price manual authoring or maintenance.

Semantic scope, and where “unchanged baseline” is exact. Exact task contracts do not certify semantic equivalence once a model continuation follows; retained-continuation replay exposes that gap. Writes, handoffs, hosted tools, undeclared effects, and guardrail evaluations at a removed turn lie outside proposition 1. The fallback claim is scoped too: every check that rejects *before* emission is exact for both integration paths, but only a staging-owning runner can restore byte-identical model input, since an adapter cannot withdraw a response the host already committed (appendix B).

8 CONCLUSION

GAC treats recurrence as candidate evidence, not permission to rewrite an agent. Across live GitHub families and a cross-repository extension, compile-or-retire saves resources while preserving exact held-out contracts; on NESTFUL and API-Bank it refuses. Judge workflow optimizers by their evidence, refusals, and fallback—not only the turns they remove.

AI USE STATEMENT

The authors used generative AI tools to restructure and copy-edit the manuscript, suggest section organization, draft and revise paper text, critique methodological exposition and internal consistency, and convert repository-grounded results into LaTeX tables. They were not used to generate experimental results or to replace manual verification. Every reported number, citation, equation, and comparison was checked against the repository artifacts and retained outputs, and the authors take responsibility for the final content.

ETHICS STATEMENT

The main risk in removing model-mediated decisions is over-trusting a compiled prefix, and thereby accelerating unsafe automation or deleting a guardrail evaluation that happened to run at the removed boundary. The method mitigates this by compiling only read-only prefixes, treating unknown effects, approvals, handoffs, and writes as hard barriers, and defaulting to retirement or fallback when evidence is insufficient. All study data are public records (GitHub repository metadata) or public benchmarks; no personal or private data were collected. The artifact is not a license to bypass human review, legal or compliance controls, or safety guardrails; unsupported regimes are part of the result, not exceptions to hide.

REPRODUCIBILITY STATEMENT

Algorithms 1 and 2 give the deployed compiler and admission rule, Algs. 3 and 4 give provenance construction and runtime dispatch, appendix A proves proposition 1, and appendix C lists every hyperparameter of the reported runs. The accompanying artifact contains source manifests, raw result files, deterministic table and figure generation, and the scripts for the live GitHub studies, the cross-repository extension, NESTFUL, and API-Bank. Before upload, these materials will be supplied as an anonymous archive with hashes and manifests preserved.

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Algorithm 3 BUILDPATG — typed argument provenance. A slot with no witness is a genuine model decision; a slot with too many witnesses is ambiguous. Both block the region rather than defaulting to the cheapest explanation.

Input: episodes $\mathcal{T}$; entry schema Θ ; transform library $\mathcal{L}$; ambiguity cap κ
Output: provenance graph G with one edge per grounded argument slot

```

1:  $G \leftarrow \emptyset$ 
2: for all episodes  $E = (z, M, e_{1:T}, y) \in \mathcal{T}$  do
3:    $\Sigma \leftarrow \{("z", \pi, v) : (\pi, v) \in \text{FLATTEN}(z), \pi \in \Theta\}$  ▷ only allowlisted entry paths are eligible
4:   for all tool calls  $c_j \in e_{1:T}$  in order do
5:     for all argument slots  $u \in \text{args}(c_j)$  do
6:        $\mathcal{H} \leftarrow \{\sigma \in \Sigma : \exists g \in \mathcal{L}, g(\sigma) = u, \text{depth}(g) \leq 2\}$ 
7:       if  $u$  is declared literal-only for  $c_j$  then
8:          $\mathcal{H} \leftarrow \{\sigma \in \mathcal{H} : g = \text{id or } g \text{ constant}\}$  ▷ never reconstruct a page index or a limit
9:       end if
10:      if  $\mathcal{H} = \emptyset$  then
11:        mark  $c_j$  MODEL-ORIGINATED; block region
12:      else if  $|\mathcal{H}| > \kappa$  then
13:        mark  $c_j$  AMBIGUOUS; block region
14:      else
15:         $G \leftarrow G \cup \{\sigma \xrightarrow{g} (c_j, u)\}$  for the minimum-description-length  $\sigma$ 
16:      end if
17:    end for
18:     $\Sigma \leftarrow \Sigma \cup \{(\text{var}(c_j), \pi, v) : (\pi, v) \in \text{FLATTEN}(o_j)\}$  ▷ a result witnesses only later slots
19:  end for
20:  for all pairs  $(e_s, e_t)$  sharing a resource where either writes it do
21:     $G \leftarrow G \cup \{e_s < e_t\}$  ▷ order edge: no reordering is ever proposed across a conflict
22:  end for
23: end for
24: return  $G$ 

```

A PROOF OF PROPOSITION 1

Fix one threshold $\eta \in \Lambda$ and write $\gamma = \delta/|\Lambda|$. Conditional on $n_\eta = m > 0$ admitted calibration groups, the number of violating groups k_η is binomial with success probability equal to the group-level selective risk r_η at that threshold, because the candidate program, guard, verifier, score, violation rule, and grid were frozen before any calibration group was observed and the original calibration groups are i.i.d. Inverting the one-sided exact binomial test (Clopper & Pearson, 1934) gives the upper bound $U_\eta = \text{Beta}^{-1}(1 - \gamma; k_\eta + 1, m - k_\eta)$ of eq. (7), which satisfies $\Pr\{r_\eta > U_\eta \mid n_\eta = m\} \leq \gamma$. When $m = 0$ the event is impossible under the convention that empty-admission thresholds receive the vacuous bound $U_\eta = 1$. Averaging over the random value of n_η therefore yields $\Pr\{r_\eta > U_\eta\} \leq \gamma$ for each fixed threshold. Applying the union bound over the finite grid gives $\Pr\{\exists \eta \in \Lambda : r_\eta > U_\eta\} \leq |\Lambda|\gamma = \delta$, i.e. simultaneous coverage with probability at least $1 - \delta$. On that event, any threshold whose upper bound is at most α also satisfies the target selective risk of eq. (4), including the coverage-maximizing threshold returned by Alg. 2. ■

Two scope conditions are worth restating. First, the statement is *per fixed candidate*: Alg. 1 searches over families and returns the first admissible one, and that outer search is not multiplicity-corrected. Second, i.i.d. admitted groups is an assumption about the calibration cohort, not a property the compiler verifies; the grouped split in Alg. 1 makes it plausible by construction but does not establish it under temporal drift.

B ADDITIONAL ALGORITHMS

Algorithm 3 constructs the typed provenance graph that decides eq. (1). Its two block conditions are asymmetric on purpose: a slot with no witness is treated as a genuine model decision and a slot with more than κ witnesses is treated as ambiguous, so the analysis never resolves an argument by picking the cheapest available explanation.

Algorithm 4 is the runtime counterpart of eq. (3). Cheap deterministic checks reject first and the calibrated gate runs only on what survives them, so the common case costs a registry lookup and

Algorithm 4 DISPATCH — boundary-time admission. Cheap deterministic checks reject first and the calibrated gate runs only on what survives them. Exactly one exit path compacts; the rest return the unmodified agent, and only a dirty post-commit failure raises an incident.

Input: entry state z ; runtime context M' ; registry $\mathcal{R}$; mode $m \in \{\text{off}, \text{shadow}, \text{live}\}$

Output: observations for the region, or FALLBACK to the baseline agent B

```

1: if  $m = \text{off}$  then
2:   return FALLBACK ▷ conformance: model-visible input must be byte-identical
3: end if
4:  $A \leftarrow \mathcal{R}.\text{RESOLVE}(M'.\text{compatibility}, M'.\text{partition})$ 
5: if  $A = \perp \vee A.\text{lifecycle} \neq \text{ACTIVE} \vee \neg \text{VERIFYSIGNATURE}(A)$  then
6:   return FALLBACK
7: end if
8: if  $H_A(z, M') \neq 1$  then
9:   return FALLBACK ▷ manifest pins, isolation keys, typed hulls, effect set
10: end if
11: if  $\text{tools}(A.P) \cap \text{ALREADYOBSERVED} \neq \emptyset$  then
12:   return FALLBACK ▷ the region already ran; its live-ins are not the fitted ones
13: end if
14: if  $m = \text{live} \wedge \exists f \in \text{tools}(A.P) : C(f).\text{quota\_attested} \wedge \text{SNAPSHOT} = \perp$  then
15:   return FALLBACK ▷ no staging owner, so a discarded attempt is not attestable
16: end if
17: if  $q_A(z) > \eta_A$  then
18:   return FALLBACK ▷ calibrated abstention on an uncertain input
19: end if
20: if  $m = \text{shadow}$  then
21:   log would-dispatch; return FALLBACK ▷ observe coverage without changing behaviour
22: end if
23:  $S \leftarrow \text{STAGE.BEGIN}(); r \leftarrow \text{INTERPRET}(A.P, z, \text{FACADE}(C))$  ▷ the facade re-checks effects at execution time
24: if  $r$  raised PreCommitError then
25:   assert  $S.\text{REVERSIBLE}();$  return FALLBACK
26: else if  $r$  raised PostCommitError then
27:   return INCIDENT ▷ an external commitment happened; do not feign rollback
28: end if
29: if  $V_A(r) \neq 1$  then
30:   discard  $r$ ; return FALLBACK ▷ live-outs, effect multiset, and call count are checked before use
31: end if
32:  $S.\text{COMMIT}(r.\text{effects});$  return  $r.\text{observations}$ 

```

a guard evaluation. Exactly one exit path compacts; five return the unmodified baseline agent, and one — a post-commit failure — raises an incident rather than feigning a rollback. “Baseline” is byte-exact only where a staging owner holds the commit boundary: the outer runner can restore byte-identical model input, whereas a model-boundary adapter cannot un-emit a response the host has already committed to conversation history.

C CONFIGURATION

All reported admission runs use risk budget $\alpha = 0.05$, confidence budget $\delta = 0.1$, and a fixed grid of $|\Lambda| = 11$ thresholds, which fixes the zero-violation sample requirement at $n_\eta \geq \log(\delta/|\Lambda|)/\log(1 - \alpha) = 91.6$, i.e. 92 admitted groups; at exactly $n_\eta = 92$ and $k_\eta = 0$ the bound is $U_\eta = 1 - (0.1/11)^{1/92} = 0.0498 \leq \alpha$. Tightening the confidence budget to $\delta = 0.05$ would raise the requirement to 106 groups, which no study in this paper reaches, so $\delta = 0.1$ is the operative setting throughout and is reported as such rather than rounded to match α . A Bonferroni repair over $m = 2$ candidate families would use $\gamma = \delta/(m|\Lambda|)$ and land on the same 106-group requirement, which is why the paper reports per-candidate certificates rather than a compiler-wide one. The grid is $\Lambda = \{.02, .05, .08, .11, .14, .17, .20, .25, .30, .40, .50\}$ and the ambiguity cap is $\kappa = 3$. Synthesis uses a 23-operator closed library at depth ≤ 2 ; region mining uses $w_{\max}$ tool calls per window and requires support across independent scenario groups. The implementation also supports a minimum-distinct-days requirement, which the single-snapshot live study does not exercise ($\text{min_days} = 1$). Two terms of eq. (5) are coarser in the implementation than the notation suggests: ρ_{eff} is approximated by the

fraction of tool names containing a namespace separator rather than read from the effect catalog, and $|F|$ is substituted by the argument-slot count of the first observed window because ranking precedes synthesis. Both appear only in a ranking heuristic — no admission decision depends on either, and the effect catalog is consulted where it is load-bearing, in qualification and in the runtime facade.

D PAIRED STATISTICS FOR THE PRIMARY FAMILIES

Per-record paired differences (compiled minus baseline) on the two families that admit a deep prefix, with 10,000-sample bootstrap intervals and two-sided Wilcoxon signed-rank tests over $n = 30$ paired held-out records:

Family	Endpoint	Paired difference (95% CI)	p
PR-outcome	Total tokens	-2196.7 [-2211.1, -2182.5]	1.7×10^{-6}
	Wall latency	-3889 ms [-4499, -3346]	1.9×10^{-9}
	Estimated cost	$-\$5.11 \times 10^{-4}$	1.7×10^{-6}
Backlog	Total tokens	-2315.0 [-2366.0, -2224.1]	1.7×10^{-6}
	Wall latency	-4428 ms [-5400, -3558]	1.9×10^{-9}
	Estimated cost	$-\$5.44 \times 10^{-4}$	1.7×10^{-6}

Provider-request differences are deterministic by construction (every pair removes the same number of turns), so no rank test is reported for them: the statistic would measure the degeneracy rather than the effect.

E ADDITIONAL EXPERIMENTAL DETAIL

Primary workflow families. The three-family result is the main paper’s strongest live evidence because it combines distinct tool vocabularies, exact task graders, balanced held-out cohorts, and provider-backed execution. The issue-type family intentionally retains a shallower two-read prefix; the newer pull-request and backlog families admit deeper pre-model compaction. This difference explains the lower resource savings on the first family and is part of the evidence boundary, not noise to average away.

Cross-repository extension. The cross-repository study is narrower than the primary workflow families. Its task uses two read-only tools and a simpler exact outcome contract, which is precisely why a fixed template can remain tied or even more efficient than the learned artifact. We keep the result in the main text because it addresses a reviewer-critical scope question—whether guarded specialization survives beyond one repository—while being explicit about the reduced task complexity.

The position invariant, empirically. An early pilot dispatched a compiled *suffix* region at the entry boundary, violating eq. (2). Because the runtime resolves artifacts before the agent has made the calls the region assumed had already happened, the pilot reordered and duplicated tool calls. The failure is a semantic mismatch between compilation-time position and resolution-time position, not a tuning error, which is why eq. (2) is a hard constraint in the deployed compiler rather than a scoring penalty.

Omitted from the main claim. The repository retains additional studies that are useful scientifically but less suitable for the main narrative:

- a natural-order issue study that exposed a factual continuation miss for a more aggressive three-read artifact,
- exploratory guarded composite synthesis and a follow-up comparison to a fair pre-model manual program,
- a bounded GEPA prompt-optimization study in which GEPA retained its seed, so the combined arm is a replication rather than evidence of interaction, and
- a portfolio pilot and controlled demonstration suite for runtime boundary coverage, including workloads where removing turns *increases* cost through prefix-cache fragmentation.

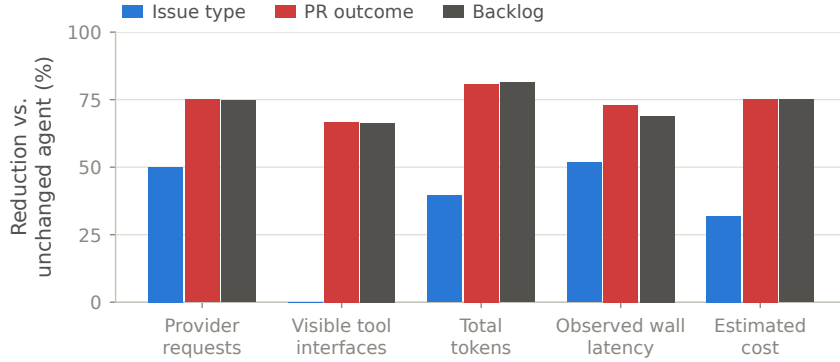


Figure 4: Per-family resource reductions behind table 2, plotted because the spread is the result: admissible prefix depth — not the optimizer — sets the savings, and the two deeper families sit at the feasibility ceiling of eq. (6).

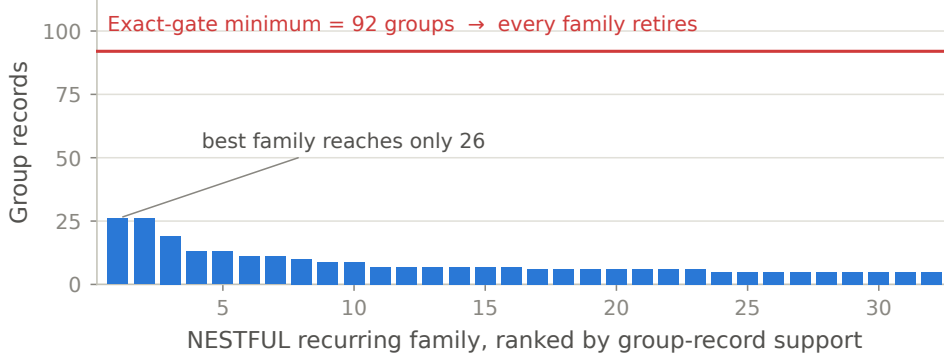


Figure 5: NESTFUL family supports against the 92-group requirement implied by eq. (7) at $\alpha = 0.05$, $\delta = 0.1$, $|\Lambda| = 11$. The largest family reaches 26 groups, so every family retires at calibration even though provenance, synthesis, and replay all succeeded.

These studies remain important to the broader artifact, but they either use a weaker risk level, address a narrower engineering question, or are explicitly exploratory.

F EXTENDED LIMITATIONS

Figure 4 plots the per-family spread behind table 2: admissible prefix depth, not the optimizer, sets the savings. The current empirical gate does not yet demonstrate a useful risk–coverage frontier at the registered 5% level. The strongest current runs are mostly all-or-none support tests (fig. 5), and the compiler-wide candidate-search problem is not yet multiplicity-corrected. Likewise, exact task contracts do not certify full semantic equivalence of the final answer, especially once a downstream model continuation is involved. Two further operational caveats belong in the record: the headline live artifact was compiled without a sandbox available to the perturbation suite, so it records `perturbations_claimed: false` rather than a completed challenge, and registry signature verification was configured off in the reported run. These are not hidden defects in the repository; they are the reasons the paper avoids stronger deployment claims.